**University of Minho**
School of Engineering

Intelligent Systems Associate Laboratory

# Dados e Aprendizagem Automática

## Interpretability and Explainability

**DAA 2025/2026**

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*Part VII*

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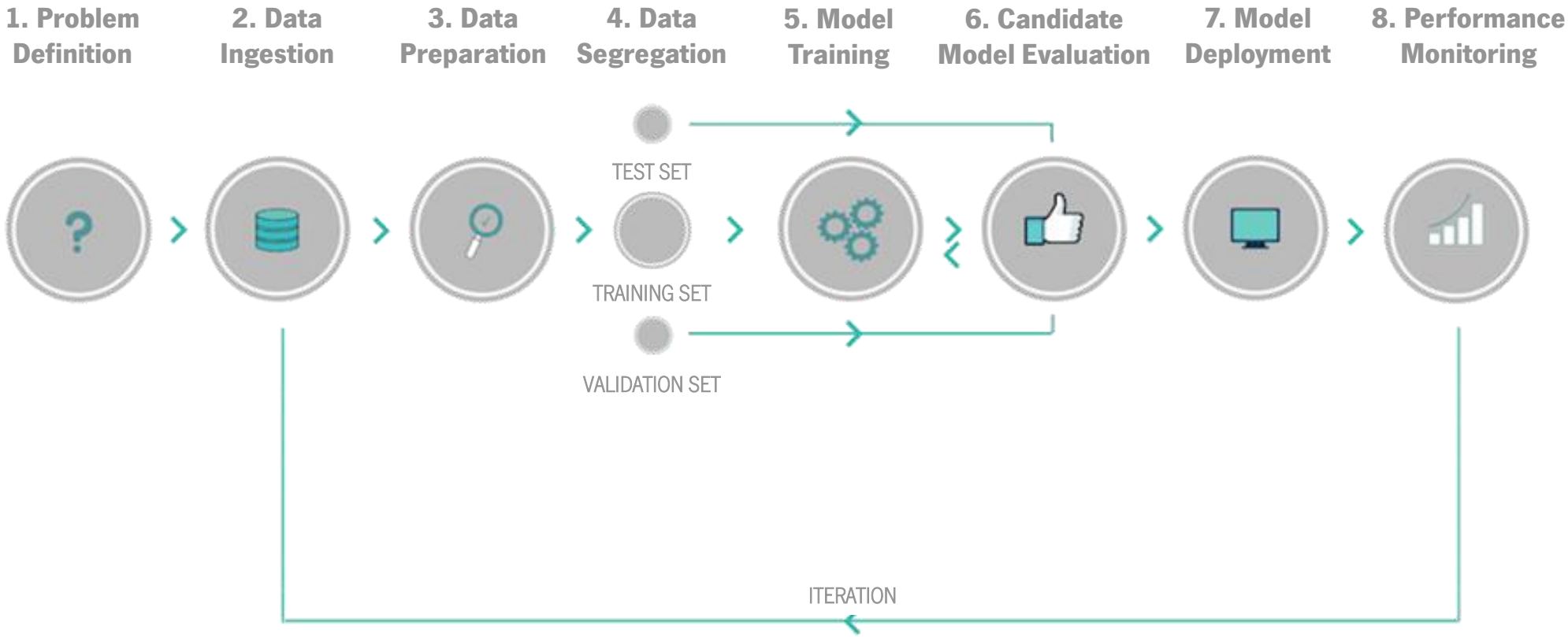
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# Interpretability and Explainability

# Interpretability and Explainability



How we can understand the cause of the ML model's decision

**WHY**

Helps to understand the behavior of the model

How to explain the model behavior *post hoc*

**HOW**

Helps understand the model's decision and its underlying mechanics and features

Both provide transparency into automated decisions, fostering trust, accountability, and the ability to debug algorithms when necessary.

# The Problem and the Data



Problem: development of a Machine Learning model that can **predict which passengers survived** the Titanic disaster.

Dataset: information regarding the **passengers** with 891 entries and 12 features, including:

- *PassengerId*
- *Survived*
- *Pclass*
- *Name*
- *Sex*
- *Age*
- *SibSp*
- *Parch*
- *Ticket*
- *Fare*
- *Cabin*
- *Embarked*

Why was it predicted that a particular passenger would or wouldn't survive?
How can we explain the model's decision?

# The Problem and the Data



Problem: development of a Machine Learning model that can **predict which passengers survived** the Titanic disaster.

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- *PassengerId*
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- *Name*
- *Sex*
- *Age*
- *SibSp*
- *Parch*
- *Ticket*
- *Fare*
- *Cabin*
- *Embarked*

You may need to install `shap` and `ipywidgets`. Use one of the following commands:

```
pip install shap
```

(alternantive) `conda install -c conda-forge shap`

```
pip install ipywidgets
```

# Feature Engineering and EDA



```
df.info()
```

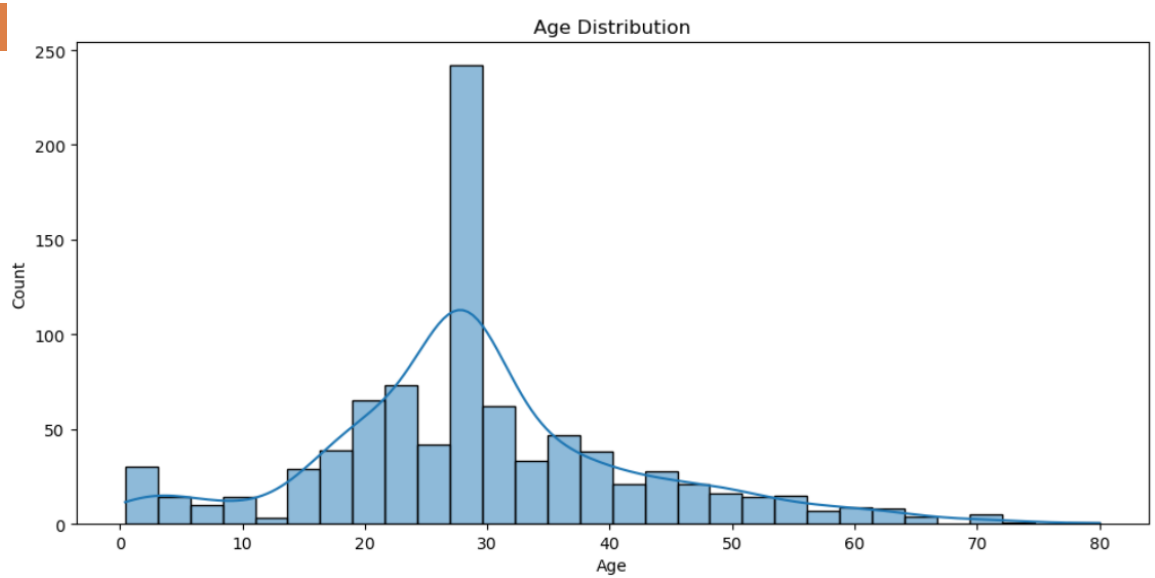
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

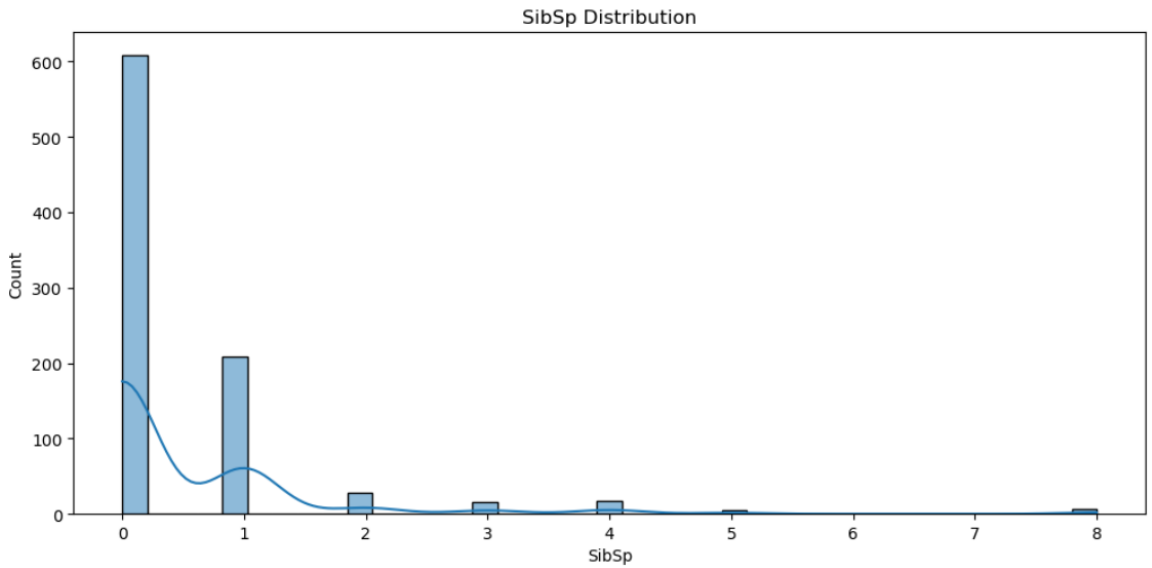
```
df.head()
```

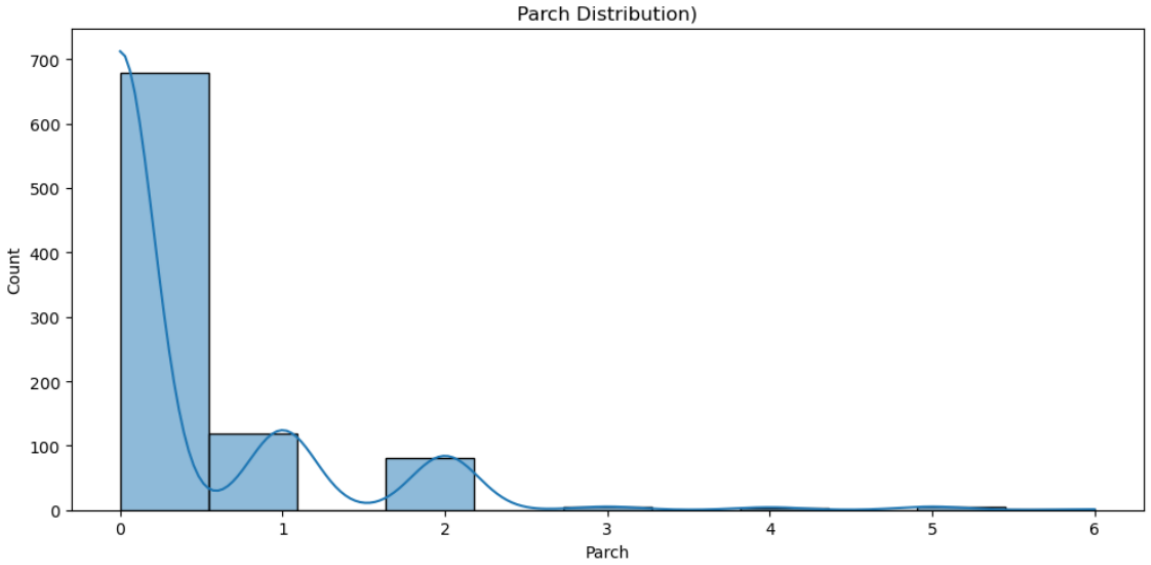
| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | NaN | S |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | NaN | S |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |

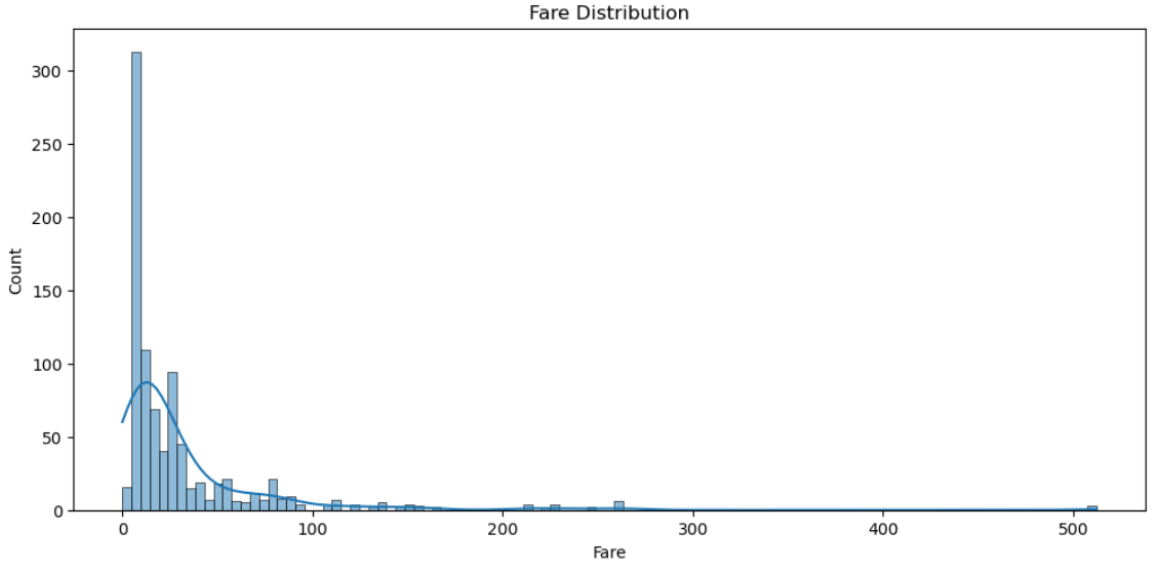
# Feature Engineering and EDA





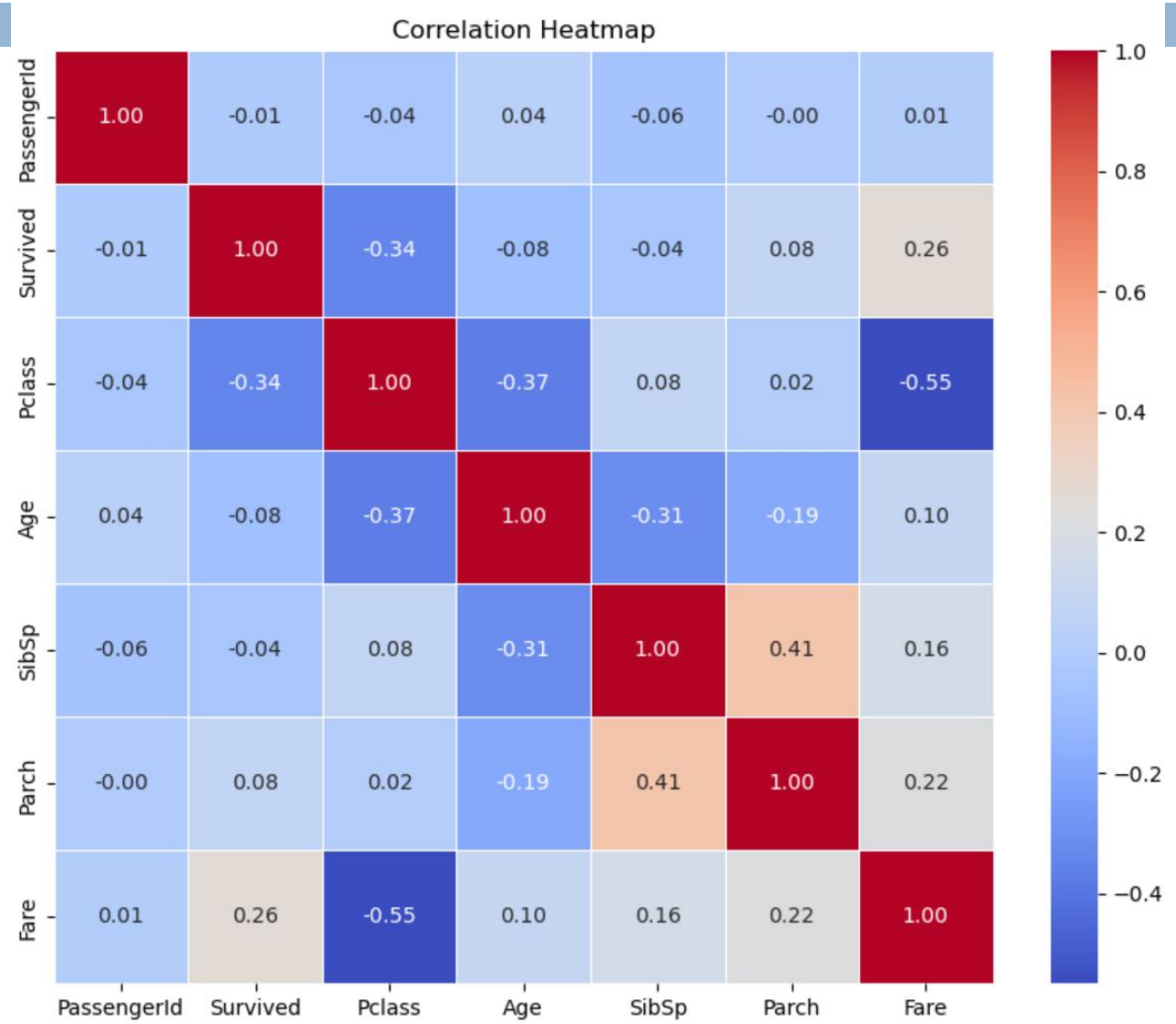



# Feature Engineering and EDA

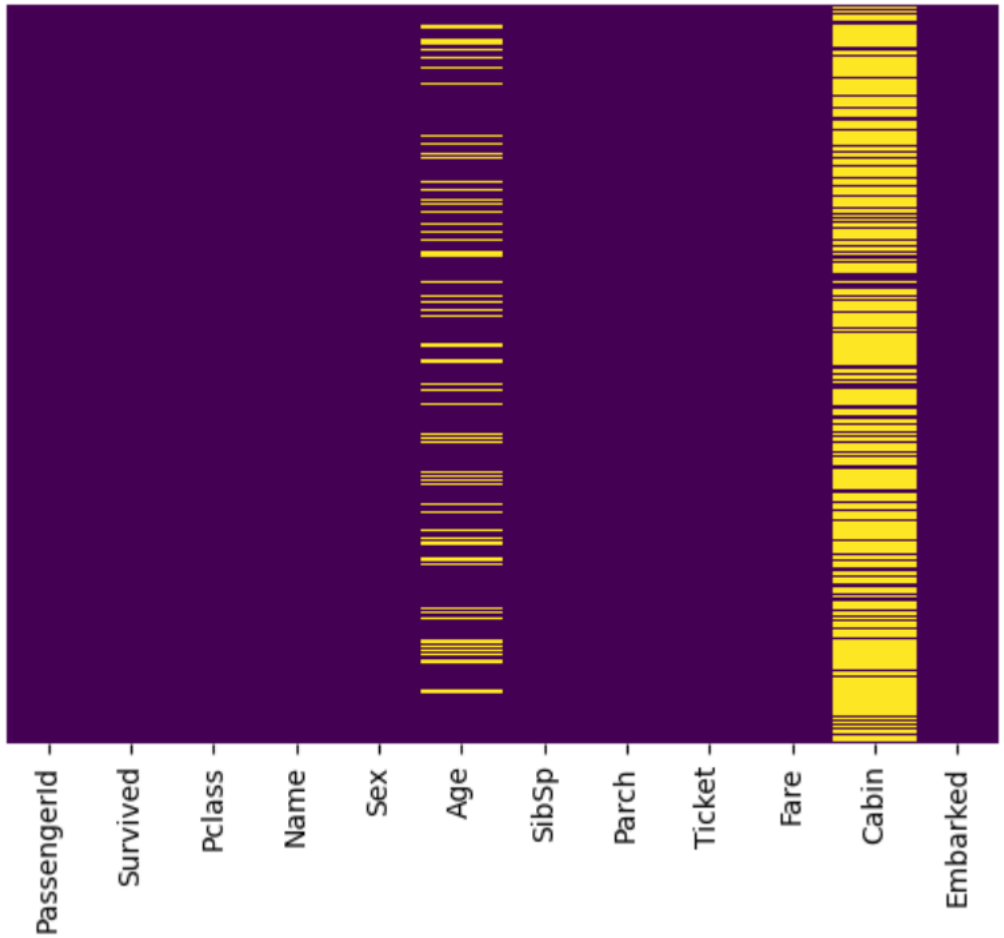




# Feature Engineering and EDA



Check for missing values.




```
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

# Feature Engineering and EDA



*Age* feature: use median to impute values.

```
def med_impute_nan(df):
    med_impute = df.copy()
    med_impute["Age"] = med_impute["Age"].fillna(med_impute["Age"].median())
    return med_impute
```

```
med_impute = med_impute_nan(df)
```

```
med_impute.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

# Feature Engineering and EDA



```
med_impute.head()
```

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | NaN | S |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | NaN | S |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0500 | NaN | S |

```
print(df['Age'].std())
print(med_impute['Age'].std())
```

```
14.526497332334044
13.019696550973194
```

# Feature Engineering and EDA

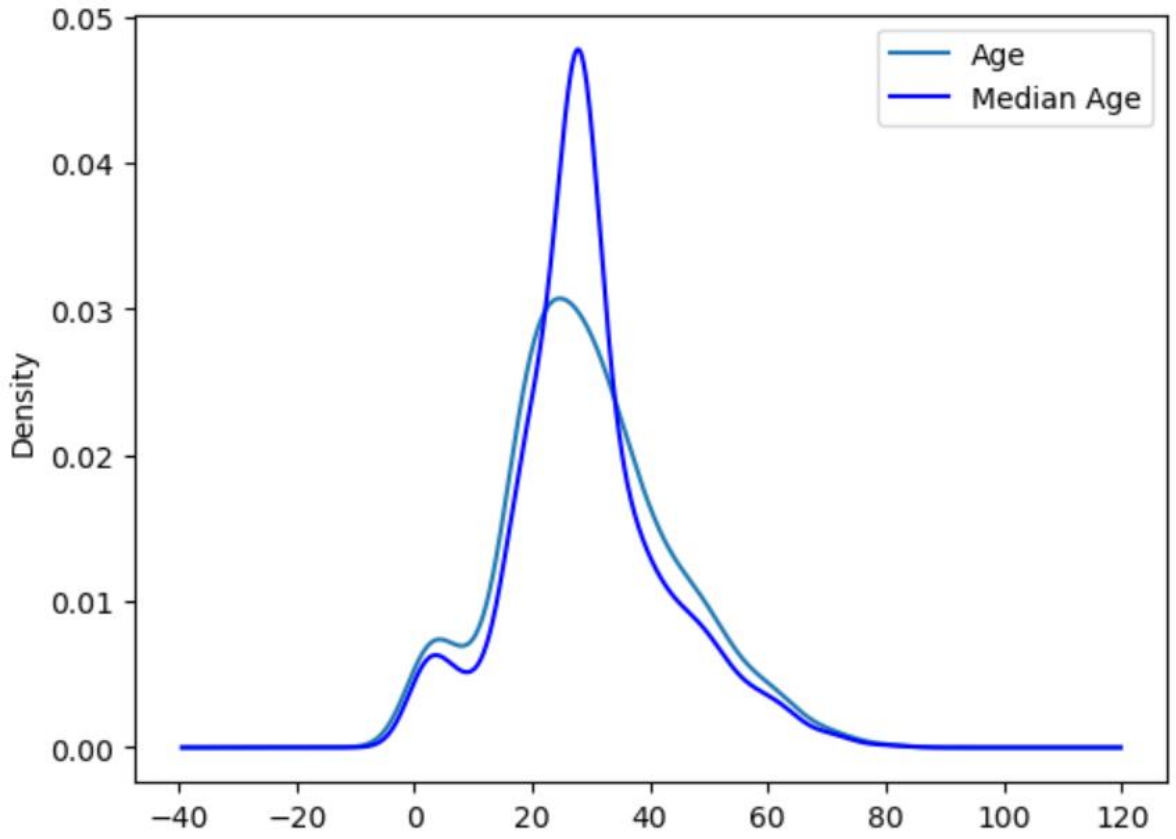


```
df = med_impute
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
fig = plt.figure()
ax = fig.add_subplot(111)
df['Age'].plot(kind='kde', ax=ax)
med_impute['Age'].plot(kind='kde', label='Median Age', ax=ax, color='blue')
lines, labels = ax.get_legend_handles_labels()
ax.legend(lines, labels, loc='best')
```

```
<matplotlib.legend.Legend at 0x1b6fd871220>
```



# Feature Engineering and EDA



*Embarked* feature: handle `NaN` values.

```
df.Embarked.value_counts()
```

```
Embarked
S    644
C    168
Q     77
Name: count, dtype: int64
```

```
emabark = df['Embarked'].dropna()
```

```
df[df['Embarked'].isnull()]
```

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **61** | 62 | 1 | 1 | Icard, Miss. Amelie | female | 38.0 | 0 | 0 | 113572 | 80.0 | B28 | NaN |
| **829** | 830 | 1 | 1 | Stone, Mrs. George Nelson (Martha Evelyn) | female | 62.0 | 0 | 0 | 113572 | 80.0 | B28 | NaN |

```
df['Embarked'].mode()
```

```
0    S
Name: Embarked, dtype: object
```

# Feature Engineering and EDA



```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
df[df['Embarked'].isnull()]
```

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|

```
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         0
dtype: int64
```

# Feature Engineering and EDA



*Cabin* feature: handle NaN values.

```
df['Cabin'].value_counts()
```

```
Cabin
B96 B98        4
G6             4
C23 C25 C27    4
C22 C26        3
F33            3
              ..
E34            1
C7             1
C54            1
E36            1
C148           1
Name: count, Length: 147, dtype: int64
```

```
df['Cabin'].mode()
```

```
0        B96 B98
1    C23 C25 C27
2             G6
Name: Cabin, dtype: object
```

```
df['Cabin'] = df['Cabin'].fillna(df['Cabin'].mode()[0])
```

# Feature Engineering and EDA



```
df.isnull().sum()
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

```
df.head()
```

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | B96 B98 | S |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | B96 B98 | S |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0500 | B96 B98 | S |

# Feature Engineering and EDA



*Sex* feature: factorise using the value 1 for male and 0 for female.

```
df['Sex'] = df['Sex'].apply(lambda x: 1 if x == 'male' else 0)
df['Sex']
```

```
0      1
1      0
2      0
3      0
4      1
      ..
886    1
887    0
888    0
889    1
890    1
Name: Sex, Length: 891, dtype: int64
```

```
df.head()
```

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | 1 | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | B96 B98 | S |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | 0 | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | 0 | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | B96 B98 | S |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | 0 | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | 1 | 35.0 | 0 | 0 | 373450 | 8.0500 | B96 B98 | S |

# Feature Engineering and EDA



*Name*, *Ticket* and *Cabin* features: remove them.

```
df.drop(columns=['Name', 'Ticket', 'Cabin'], inplace=True)
```

```
df.head()
```

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Fare | Embarked |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 22.0 | 1 | 0 | 7.2500 | S |
| **1** | 2 | 1 | 1 | 0 | 38.0 | 1 | 0 | 71.2833 | C |
| **2** | 3 | 1 | 3 | 0 | 26.0 | 0 | 0 | 7.9250 | S |
| **3** | 4 | 1 | 1 | 0 | 35.0 | 1 | 0 | 53.1000 | S |
| **4** | 5 | 0 | 3 | 1 | 35.0 | 0 | 0 | 8.0500 | S |

# Feature Engineering and EDA



*Embarked* feature: encode the values `S`, `C`, and `Q` using `LabelEncoder`.

```
print(df["Embarked"].value_counts())
```

```
Embarked
S    646
C    168
Q     77
Name: count, dtype: int64
```

```
from sklearn.preprocessing import LabelEncoder
cols = ['Embarked']
le = LabelEncoder()

for col in cols:
    df[col] = le.fit_transform(df[col])
df.head()
```

| | PassengerId | Survived | Pclass | Sex | Age | SibSp | Parch | Fare | Embarked |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | 1 | 22.0 | 1 | 0 | 7.2500 | 2 |
| **1** | 2 | 1 | 1 | 0 | 38.0 | 1 | 0 | 71.2833 | 0 |
| **2** | 3 | 1 | 3 | 0 | 26.0 | 0 | 0 | 7.9250 | 2 |
| **3** | 4 | 1 | 1 | 0 | 35.0 | 1 | 0 | 53.1000 | 2 |
| **4** | 5 | 0 | 3 | 1 | 35.0 | 0 | 0 | 8.0500 | 2 |

# Feature Engineering and EDA



Correlation analysis of the features.

```
correlation_matrix = df.corr()
correlation_with_target = correlation_matrix['Survived'].sort_values(ascending=False)
print("Correlation with target (Survived):\n", correlation_with_target)
```

```
Correlation with target (Survived):
 Survived       1.000000
Fare            0.257307
Parch           0.081629
PassengerId    -0.005007
SibSp          -0.035322
Age            -0.064910
Embarked       -0.167675
Pclass         -0.338481
Sex            -0.543351
Name: Survived, dtype: float64
```

In this case, the features *Sex*, *Pclass*, and *Fare* have the highest absolute correlation values with *Survived*, suggesting that they can be useful for prediction.

# Feature Engineering and EDA



Save the new data frame to a new file.

```
# Convert data to DataFrame
t = pd.DataFrame(df)

# Specify the CSV file name
filename = "titanic_ds.csv"

# Save to CSV
t.to_csv(filename, index=False, encoding='utf-8')
```

# Random Forest Classifier



Split the data.

```
X = df.drop('Survived', axis=1)
y = df['Survived']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 2022)
```

Train the model.

```
rf_model = RandomForestClassifier()
```

```
rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier()
```

# Random Forest Classifier



Obtain the model's accuracy.

```
rf_score = rf_model.score(X_test, y_test)
print("Accuracy: %.2f%%" % (rf_score * 100))
```

```
Accuracy: 82.84%
```

Save the predictions to a file.

```
op_rf = rf_model.predict(X_test)
```

```
op = pd.DataFrame(X_test['PassengerId'])
op['Survived'] = op_rf
op.to_csv("submission.csv", index=False)
```

# Feature Importance



- It measures the contribution of each feature to the model's predictive performance, revealing which attributes have the greatest impact on the results.

- **Global** vs. **Local** Feature Importance

| Feature | Global Importance | Local Importance |
|---|---|---|
| *Scope* | It explains the behaviour of the entire model across all predictions. | It explains why a specific prediction was made for a particular data point. |
| *Use Case* | Identifying the most influential features for the model as a whole. | Debugging a single prediction or justifying a decision to an individual. |
| *Example* | A partial dependence plot showing that the average price of a car tends to decrease as the number of repairs increases. | Using LIME or Shapley values to demonstrate that a loan application was rejected due to high debt and a low credit score. |

# Feature Importance



- **Model-Agnostic** vs. **Model-Specific**

| Feature | Model-Agnostic | Model-Specific |
|---|---|---|
| *Methodology* | The model is treated as a "black box" and analysed to see how the output changes when the input features are altered. | It leverages the internal structure and mechanics of the model itself. |
| *Flexibility* | It can be applied to any machine learning model, regardless of its architecture. | Often more computationally efficient for the specific model it is designed for. |
| *Examples* | Global: Feature Permutation Importance (FPI).<br>Local: LIME and Shapley values. | Global: Gini importance in random forests.<br>Local: Analysing a specific neuron's activations in a neural network. |

- There are several techniques: permutation importance, tree-based importance scores, SHAP values, among others.
- Feature importance is fundamental for optimisation and model refinement, guiding the selection of relevant features to improve predictive accuracy and model efficiency.

# Random Forest Feature Importance



**Feature importance** analysis reveals how factors such as age, gender, and class impact survival rates in the Titanic dataset.

Use the `time` class to evaluate each FI type.

```
start_time = time.time()

rf_importances = rf_model.feature_importances_
std = np.std([tree.feature_importances_ for tree in rf_model.estimators_], axis=0)

elapsed_time = time.time() - start_time

print(f"Elapsed time to compute the importances: {elapsed_time:.3f} seconds")
```

```
Elapsed time to compute the importances: 0.023 seconds
```

Obtain the feature importance values.

```
print("Random Forest Feature Importances:\n", rf_importances)
```

```
Random Forest Feature Importances:
 [0.1831734  0.09478849 0.22510325 0.17173298 0.04304862 0.04527684
 0.20599605 0.03088036]
```

What do these values represent?

# Feature Importance based on Mean Decrease in Impurity (MDI)



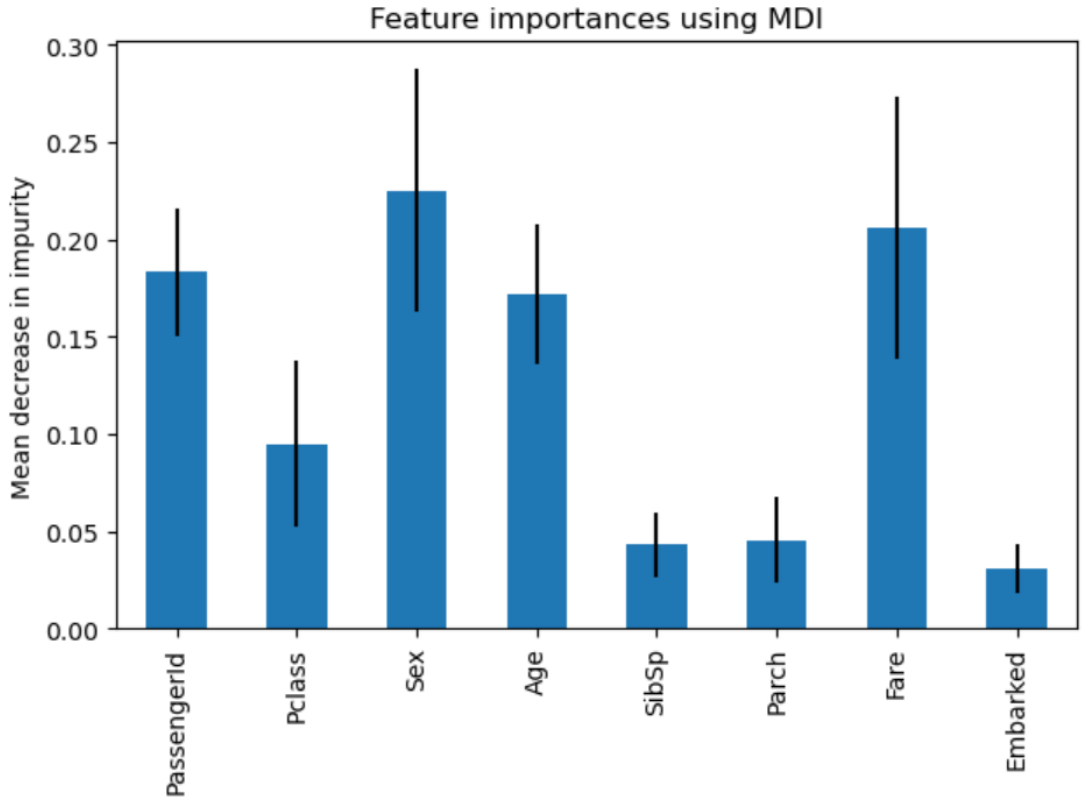
```
start_time = time.time()

mdi_importances = pd.Series(rf_model.feature_importances_, index=X_test.columns)

elapsed_time = time.time() - start_time

print(f"Elapsed time to compute the importances: {elapsed_time:.3f} seconds")
```

```
Elapsed time to compute the importances: 0.013 seconds
```

```
print("Feature importances using MDI:\n", mdi_importances)
```

```
Feature importances using MDI:
 PassengerId    0.183173
Pclass          0.094788
Sex             0.225103
Age             0.171733
SibSp           0.043049
Parch           0.045277
Fare            0.205996
Embarked        0.030880
dtype: float64
```

```
fig, ax = plt.subplots()
mdi_importances.plot.bar(yerr=std, ax=ax)
ax.set_title("Feature importances using MDI")
ax.set_ylabel("Mean decrease in impurity")
fig.tight_layout()
```



# Feature Importance based on Permutation Importance



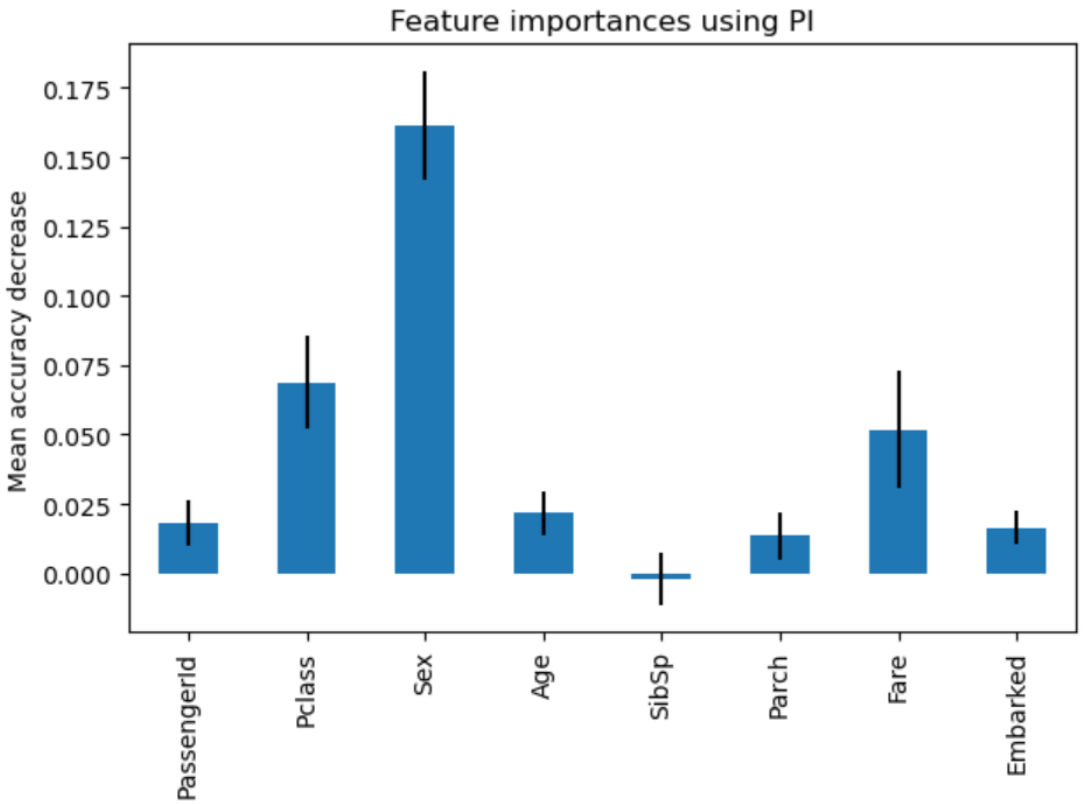
```
start_time = time.time()

result = permutation_importance(rf_model, X_test, y_test, n_repeats=10, random_state=42, n_jobs=2)

elapsed_time = time.time() - start_time
print(f"Elapsed time to compute the importances: {elapsed_time:.3f} seconds")
```

```
Elapsed time to compute the importances: 1.671 seconds
```

```
p_importances = pd.Series(result.importances_mean, index=X_test.columns)
print("Feature importances using PI:\n", p_importances)
```

```
Feature importances using PI:
 PassengerId    0.017910
Pclass          0.068657
Sex             0.161567
Age             0.021642
SibSp          -0.002239
Parch           0.013433
Fare            0.051866
Embarked        0.016418
dtype: float64
```

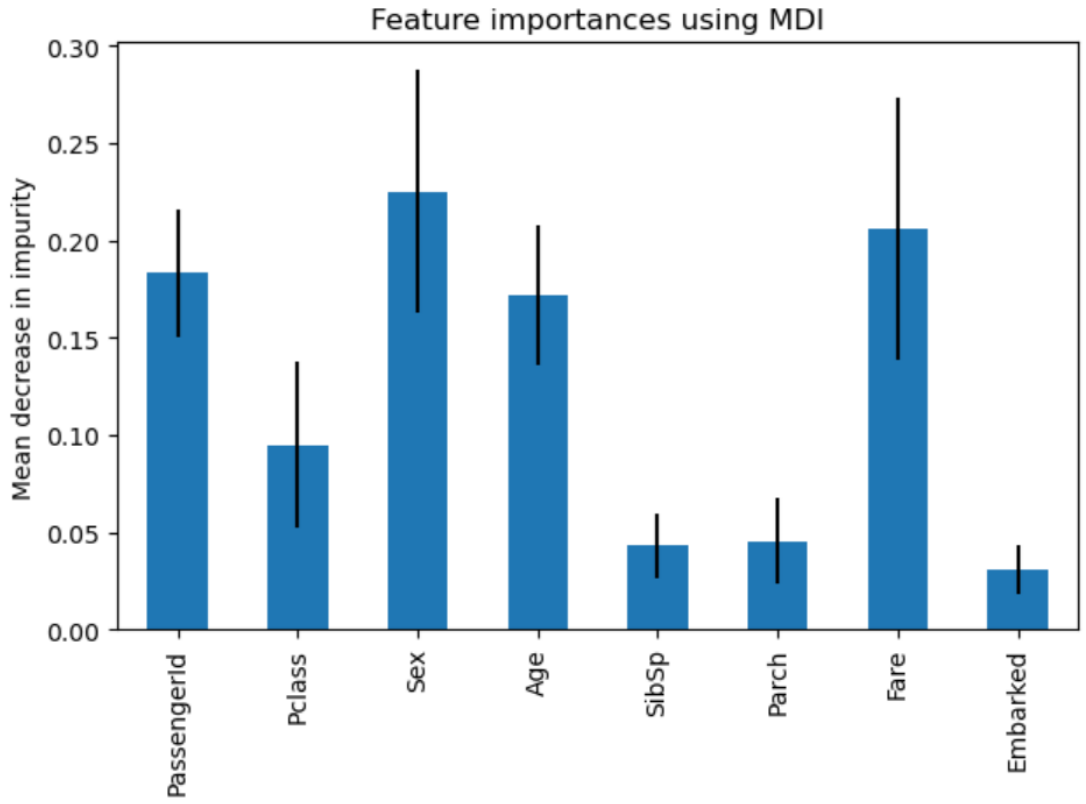
```
fig, ax = plt.subplots()
p_importances.plot.bar(yerr=result.importances_std, ax=ax)
ax.set_title("Feature importances using PI")
ax.set_ylabel("Mean accuracy decrease")
fig.tight_layout()
plt.show()
```



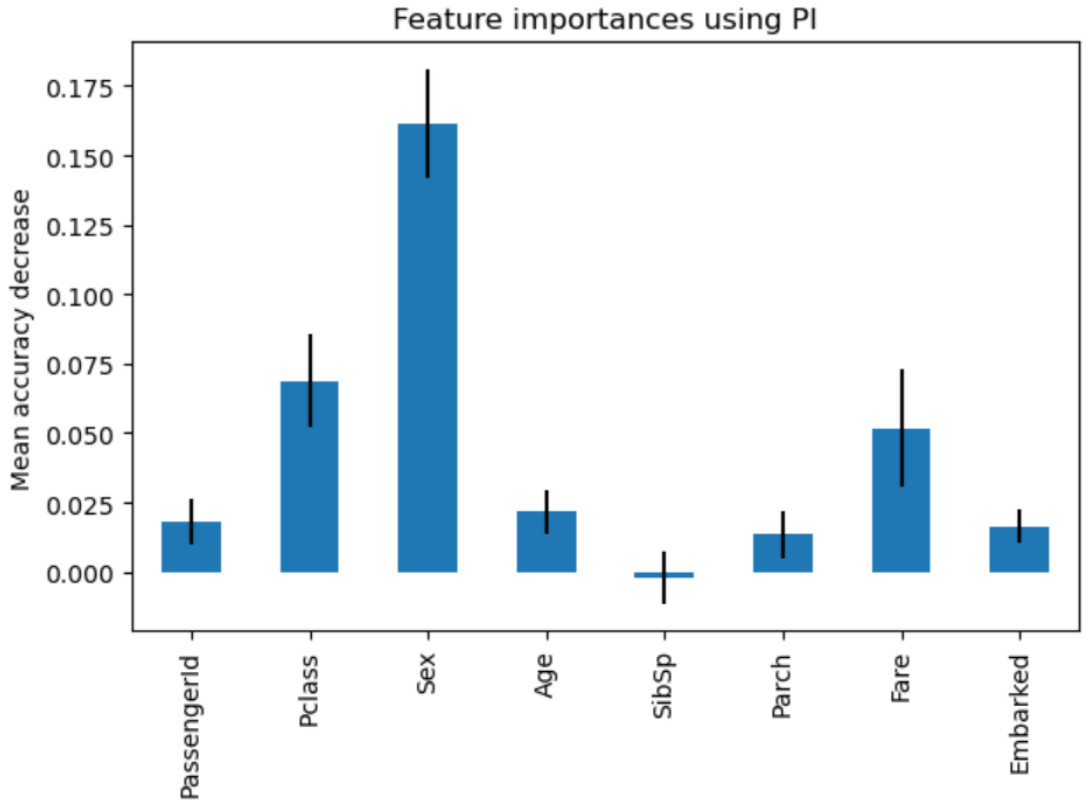
# Feature Importance



Which features are more important?




Model-specific




Model-agnostic

# SHAP (SHapley Additive exPlanations) Analysis



- Based on game theory, SHAP values fairly distribute importance among features and provide a unified framework for interpreting the output of any ML model.
- By representing each feature's contribution to a model's output, SHAP values improve our ability to analyse and validate the decision-making process of any predictive model.
- They provide accurate and consistent explanations for both global and local predictions.

- **Global interpretation**: the overall importance of features across all predictions.
- **Local interpretation**: why a specific passenger was predicted to survive or not.

- **Benefits:**
  - Broader view of feature contributions by attributing influence based on the distribution of feature values rather than using traditional methods such as simple averages or counts;
  - Fairness and comprehensiveness of SHAP scores provide better insight, help mitigate bias and ensure that the effects of interactions between features are effectively captured.

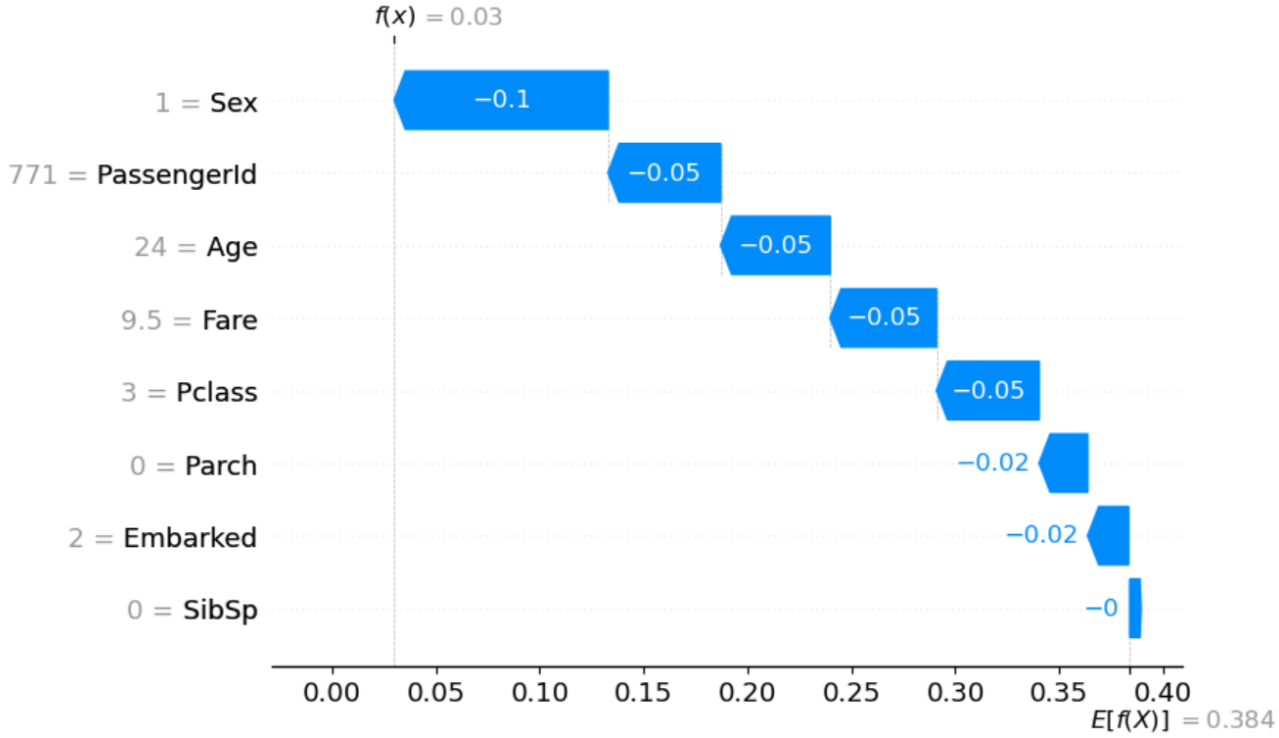
# SHAP Analysis – Local Interpretation



The Titanic dataset can be explained using **local interpretability**. Let's understand why a specific passenger (ID = 0) didn't survive.

```
no = 0
if rf_model.predict(np.expand_dims(X_test.iloc[no],axis=0))[0] == 1:
    print("The passenger survived")
else:
    print("The passenger did not survive")
shap.plots.waterfall(shap_values[no,:,1])
```

```
The passenger did not survive
```



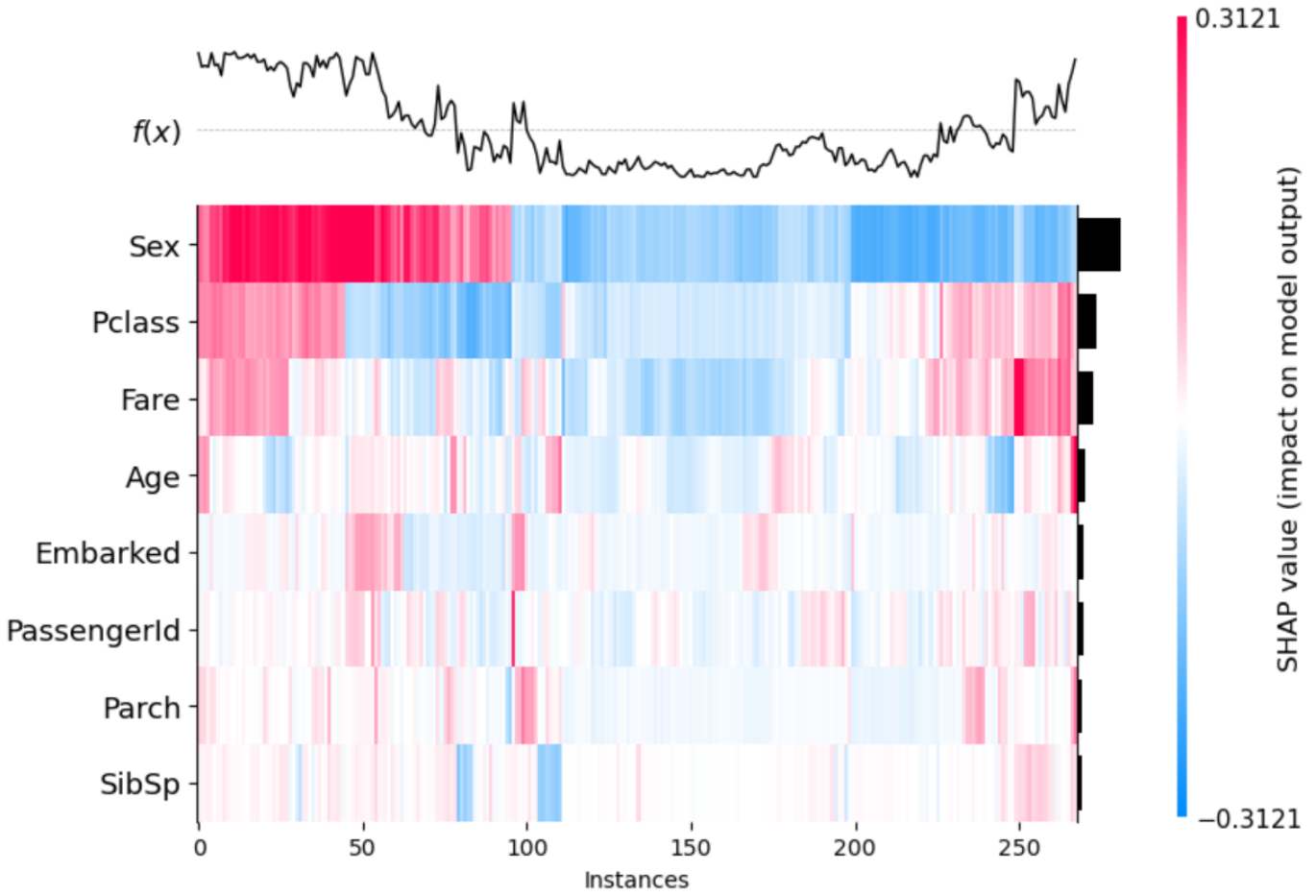

Contribution of each feature to the survival of the passenger

- Sex = 1 – male
- PassengerID = 771 – one of the last to board the ship
- Age = 24 –young age
- Fare = 9.5 – low fare
- Pclass = 3 – 3rd class
- Parch = 0 – no parents/children aboard
- Embarked = 2 – port of embarkation 2
- SibSp = 0 – no siblings/spouse aboard

# SHAP Analysis – Global Interpretation



```
shap.plots.heatmap(shap_values[:,:,1])
```



# SHAP Analysis – Feature Wise

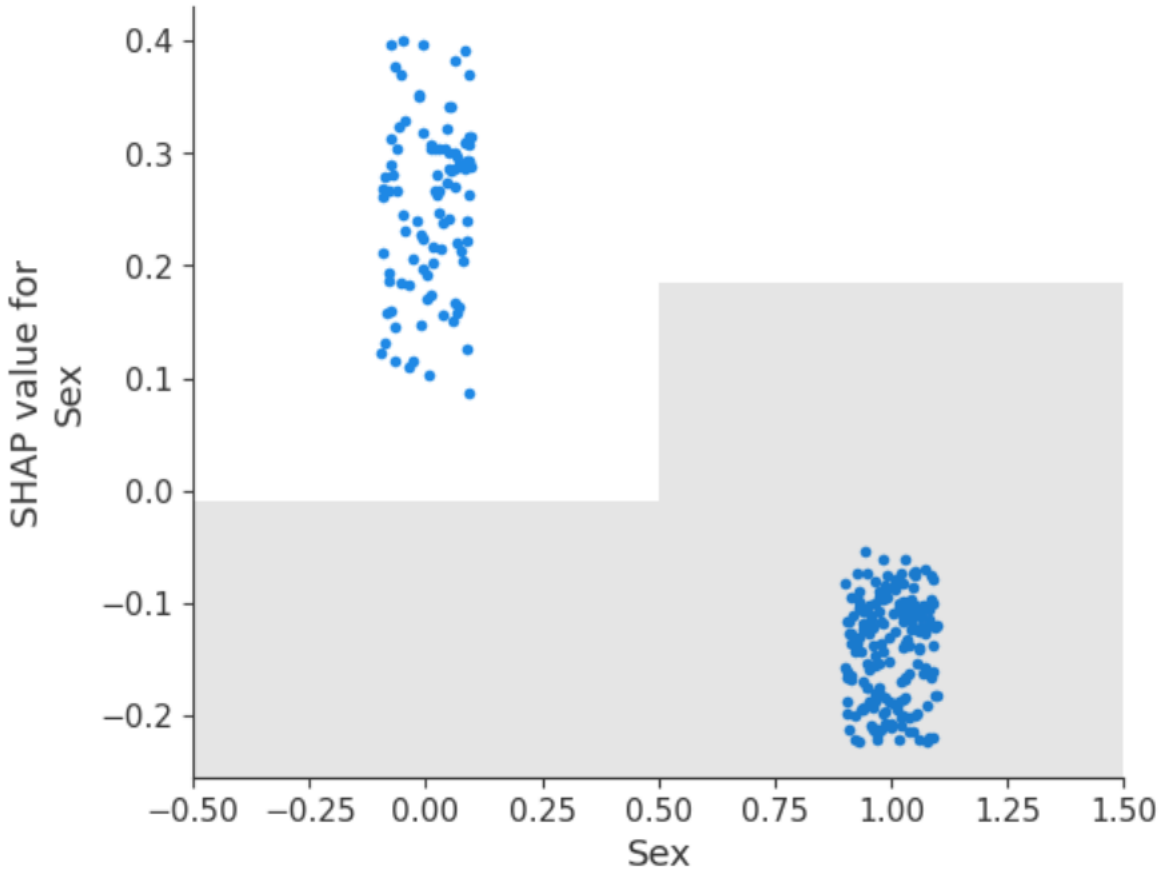


This shows how each feature contributed to the overall model prediction.

**Sex**

- Being a male (1) reduced the chances of survival.

```
shap.plots.scatter(shap_values[:,"Sex",1])
```



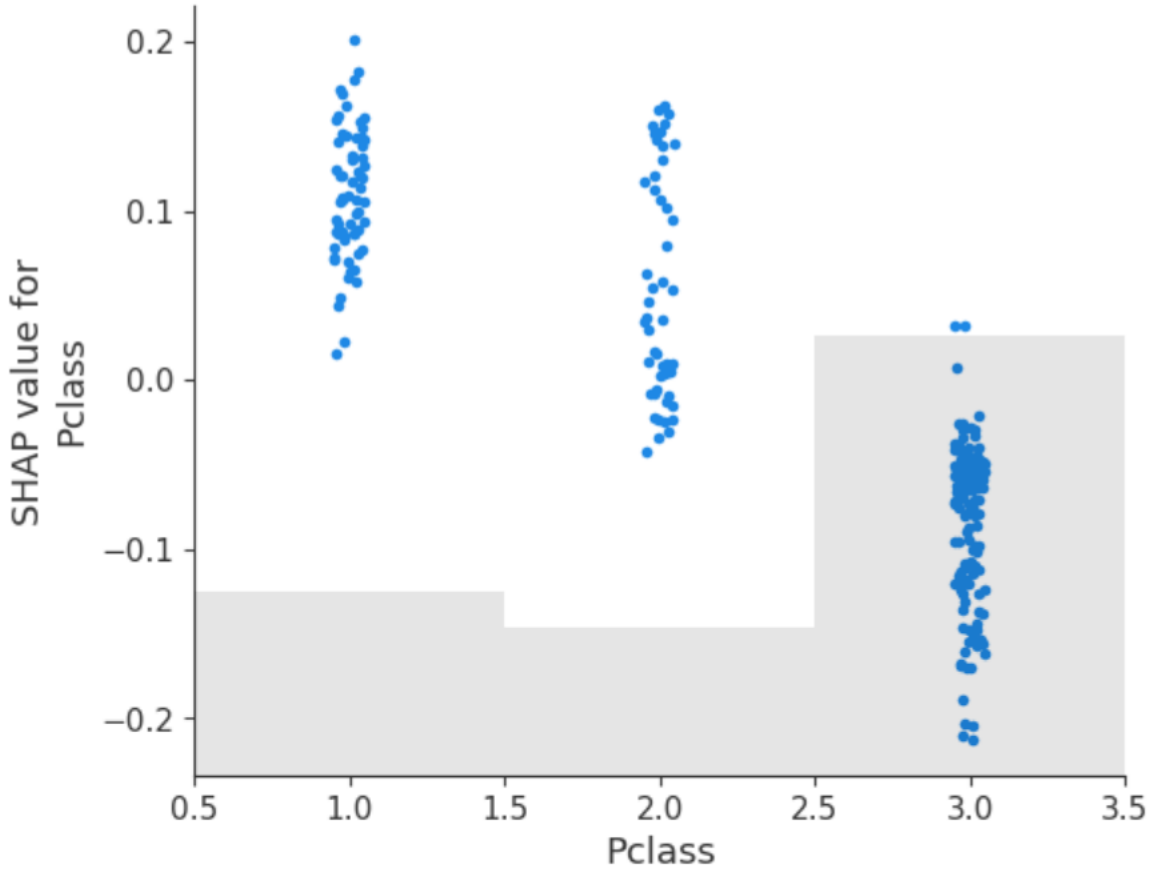
# SHAP Analysis – Feature Wise



**Pclass**

- Being in the 3rd class (3) reduced the chances of survival.
- Being in the 2nd class (2) contributed slightly positively to survival.

```
shap.plots.scatter(shap_values[:,"Pclass",1])
```



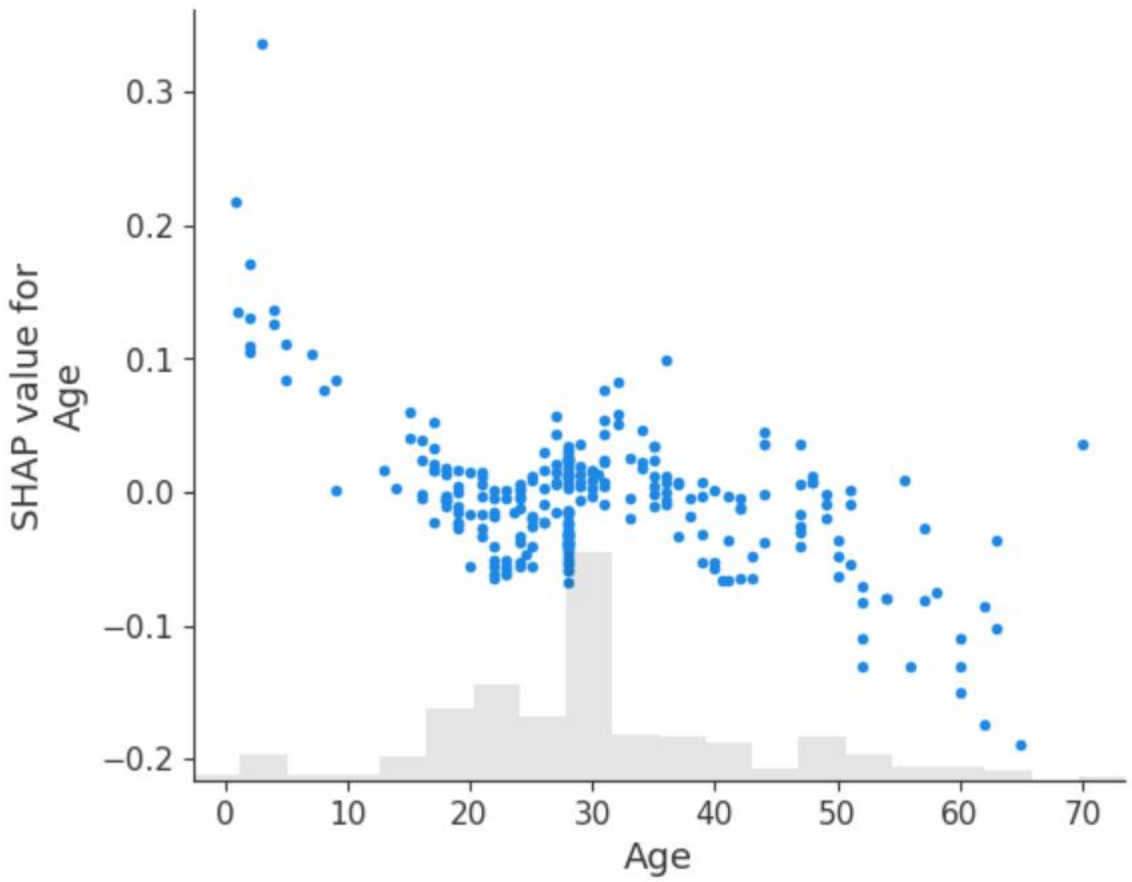
# SHAP Analysis – Feature Wise



**Age**

- Being below 10 years old increased the chances of survival, while being over 50 years old decreased them.

```
shap.plots.scatter(shap_values[:,"Age",1])
```



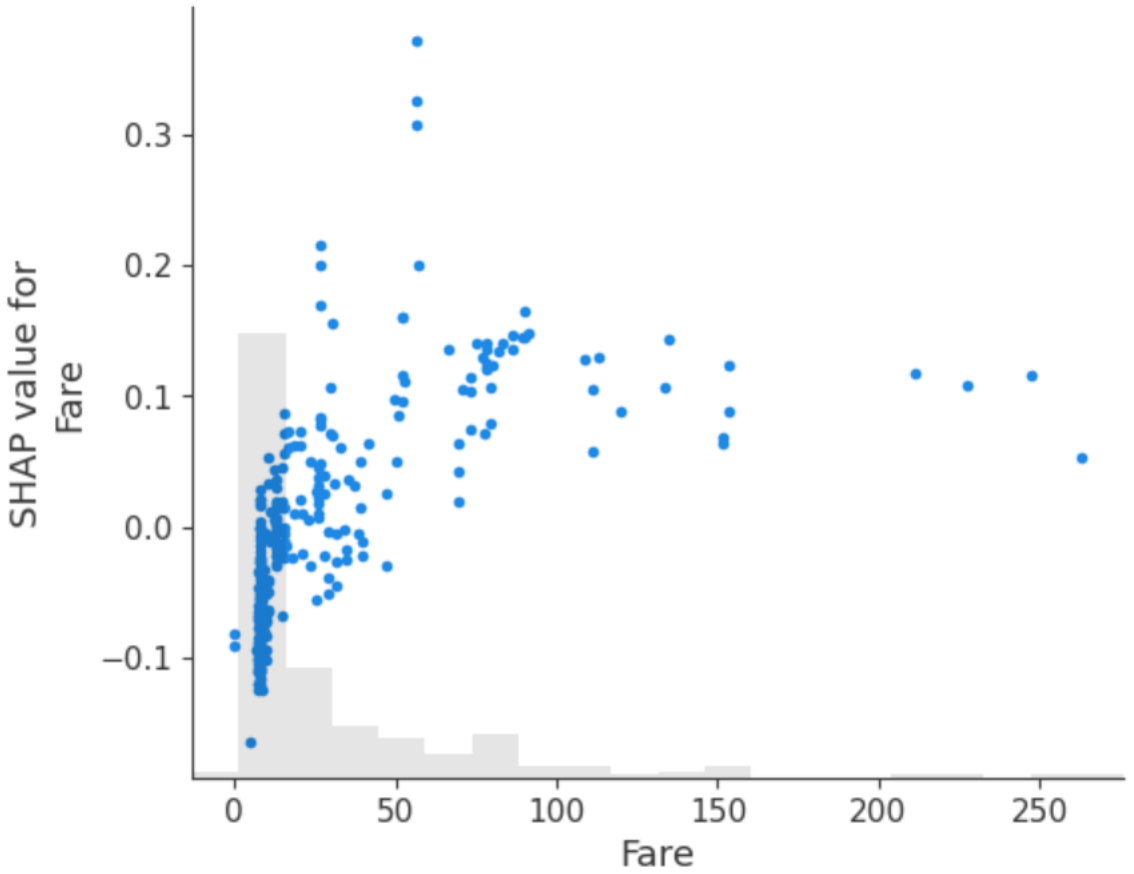
# SHAP Analysis – Feature Wise



**Fare**

- A low fare had a negative impact on survival, whereas a high fare (above 70) had a positive impact.

```
shap.plots.scatter(shap_values[:,"Fare",1])
```



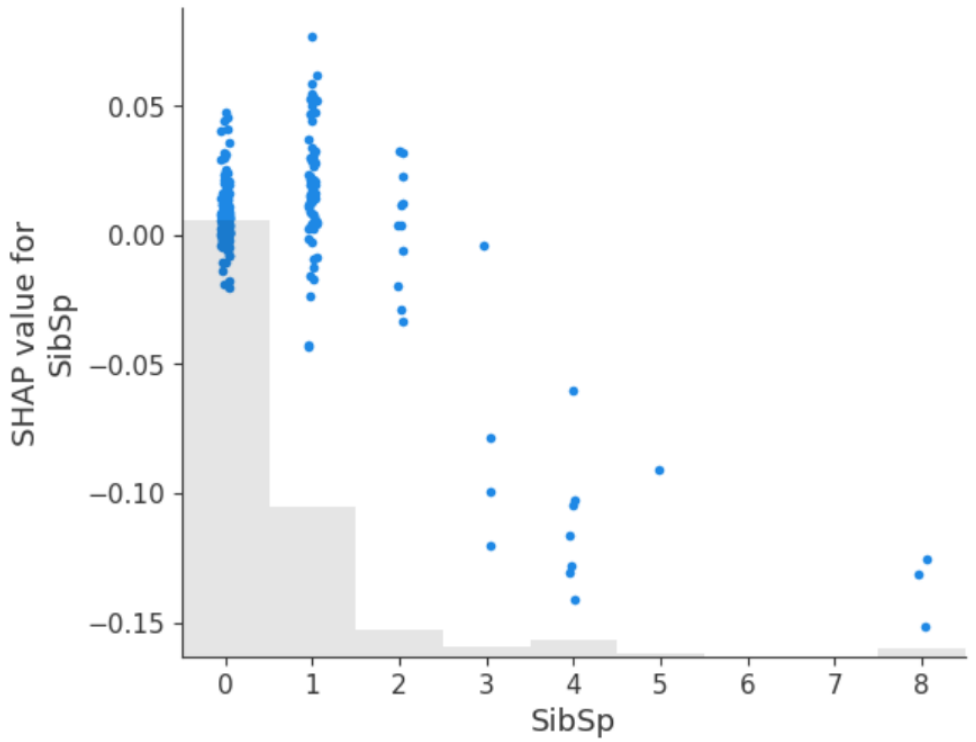
# SHAP Analysis – Feature Wise



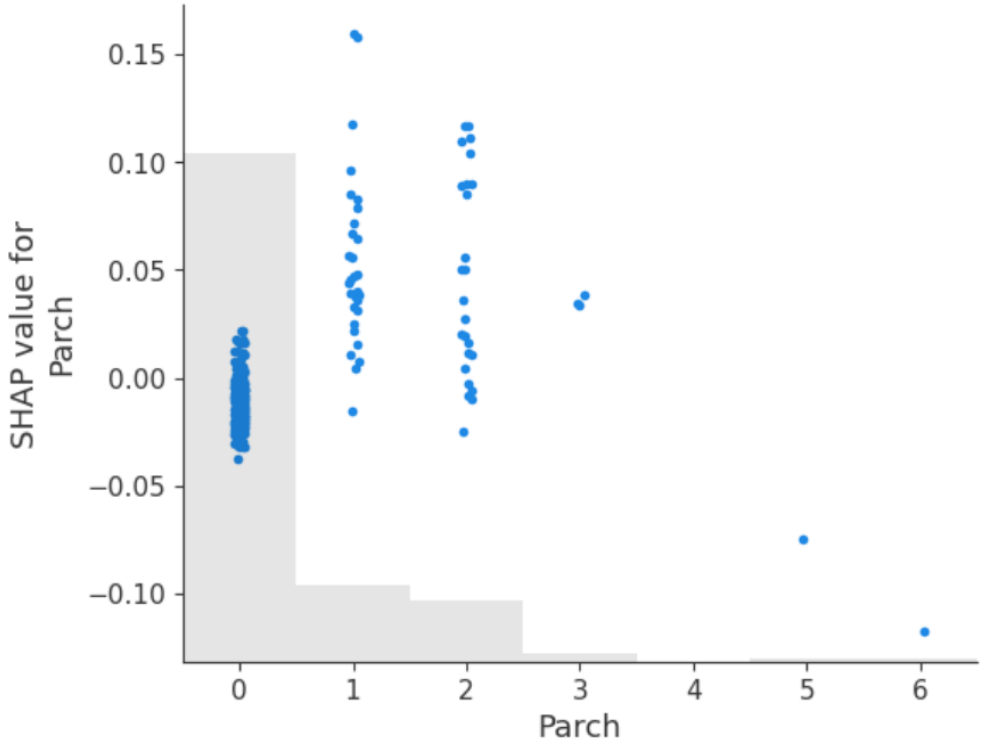
**SibSp** and **Parch**

- A higher sibling/spouse relationship had negative impact on survival.
- Having 0 or 1 sibling/spouse had positive impact on survival.
- Having 1 or 2 parents/children had a slightly positive impact on survival.

```
shap.plots.scatter(shap_values[:,"SibSp",1])
```




```
shap.plots.scatter(shap_values[:,"Parch",1])
```



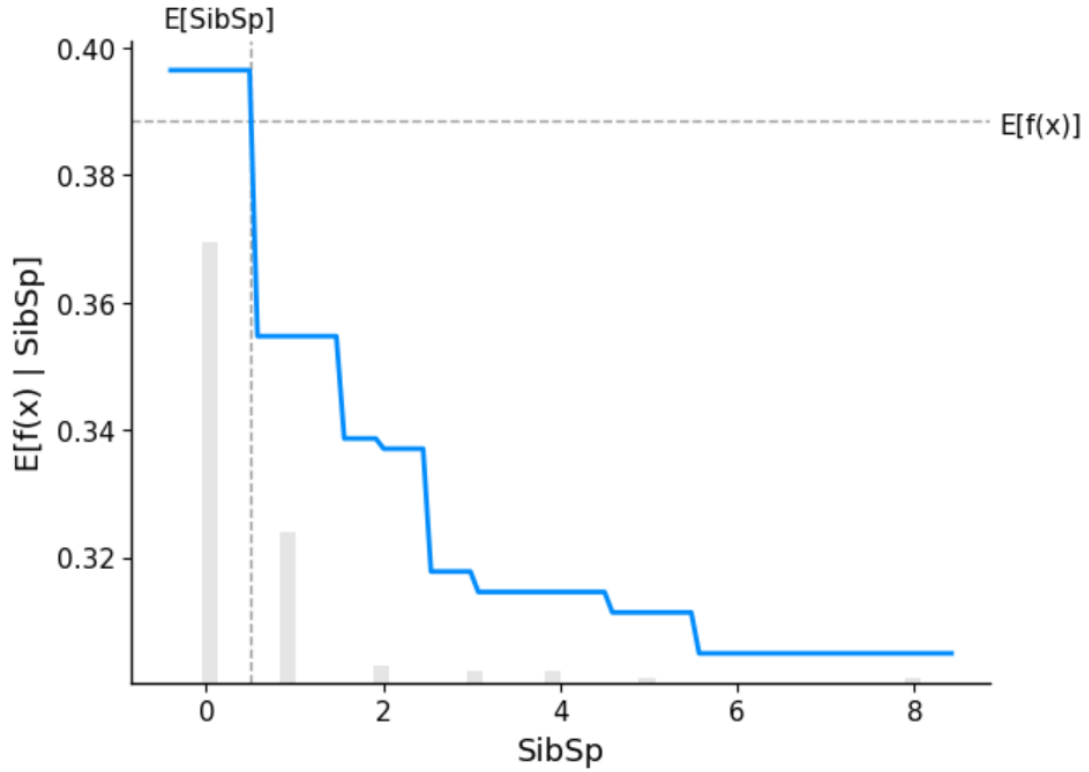
# SHAP Analysis – Feature Wise



**Global interpretability** is essential for understanding the behaviour of the entire model.

By using **Partial Dependence Plots (PDPs)** with the Titanic dataset, we can visualise the effect that changes in *SibSp* have on the model's predictions.

```
shap.partial_dependence_plot("SibSp", rf_model.predict, X_train, ice=False, model_expected_value=True, feature_expected_value=True,)
```



# SHAP Analysis – Linear Regression



Train the model.

```
lm = linear_model.LinearRegression()
lm.fit(X_train, y_train)
```

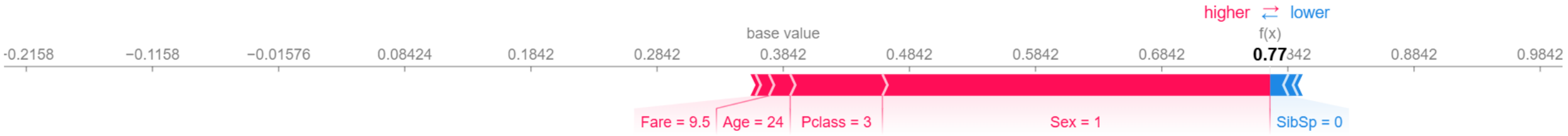
LinearRegression

LinearRegression()

```
explainer = shap.LinearExplainer(lm, X_train, feature_perturbation="interventional")
shap_values = explainer.shap_values(X_train)
```

SHAP values for one passenger.

```
shap.force_plot(explainer.expected_value, shap_values[0,:], X_train.iloc[0,:])
```



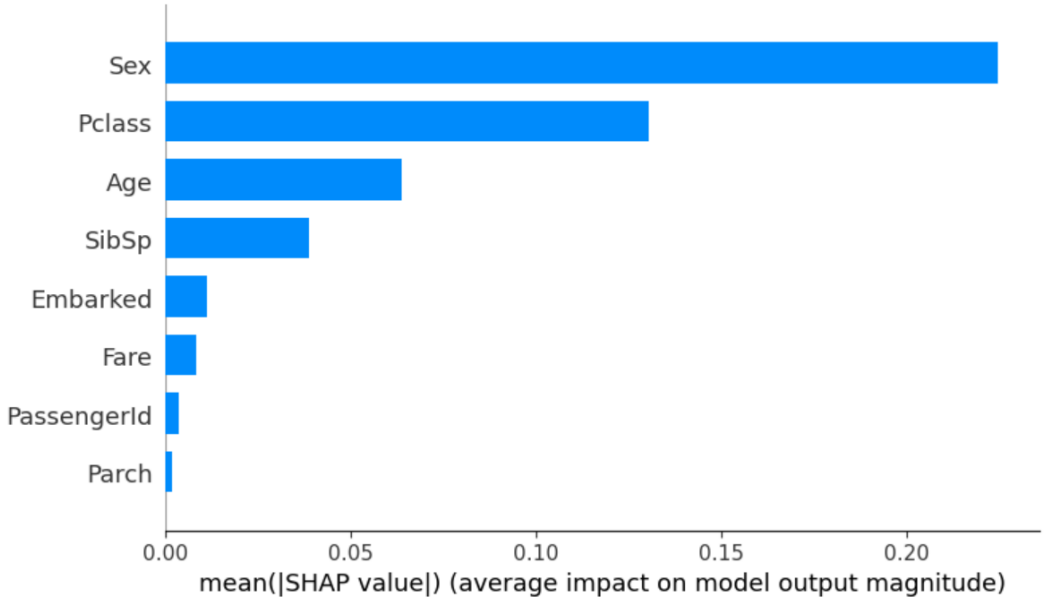
# SHAP Analysis – Linear Regression



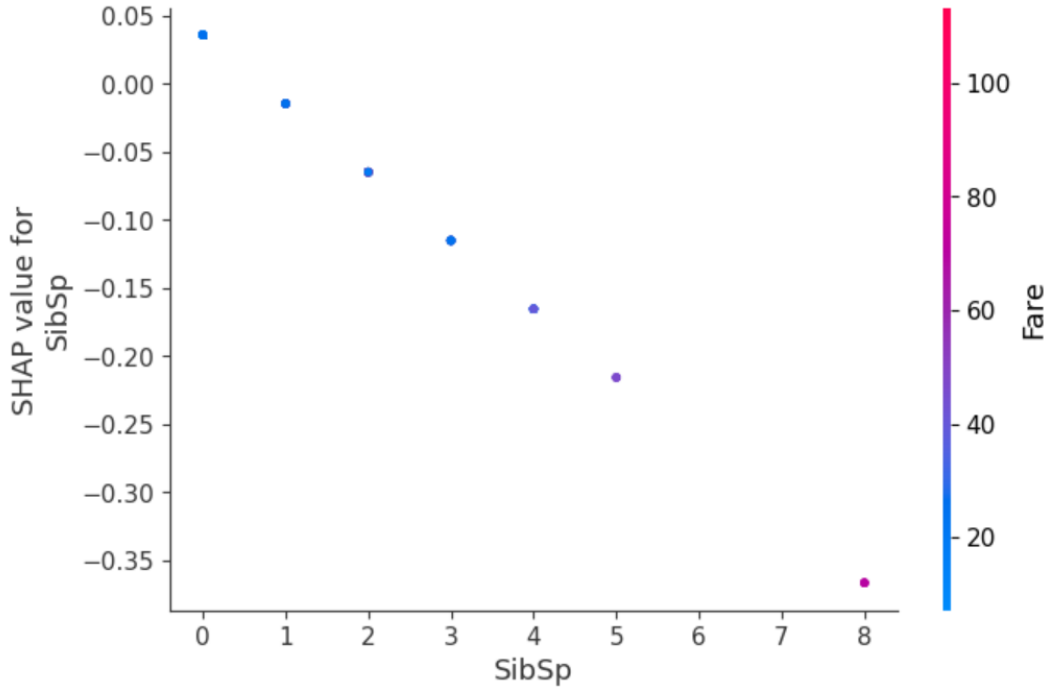
SHAP values for all training data.

```
explainer_shap = shap.LinearExplainer(model=lm, masker=X_train)
shap_values = explainer_shap.shap_values(X_train)
```

```
shap.summary_plot(shap_values, X_train, plot_type="bar")
```




```
shap.dependence_plot("SibSp", shap_values, X_train)
```

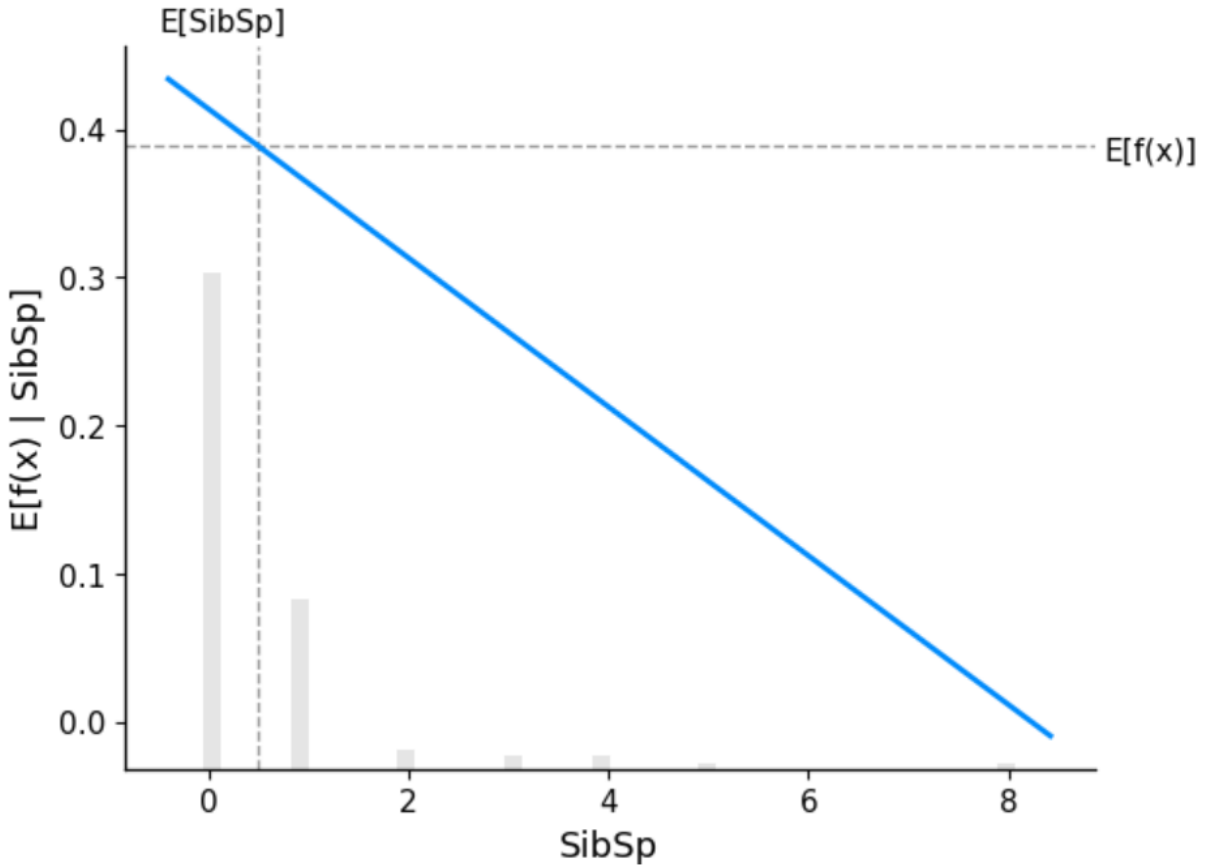


# SHAP Analysis – Feature Wise



**SibSp** – Partial Dependence Plot

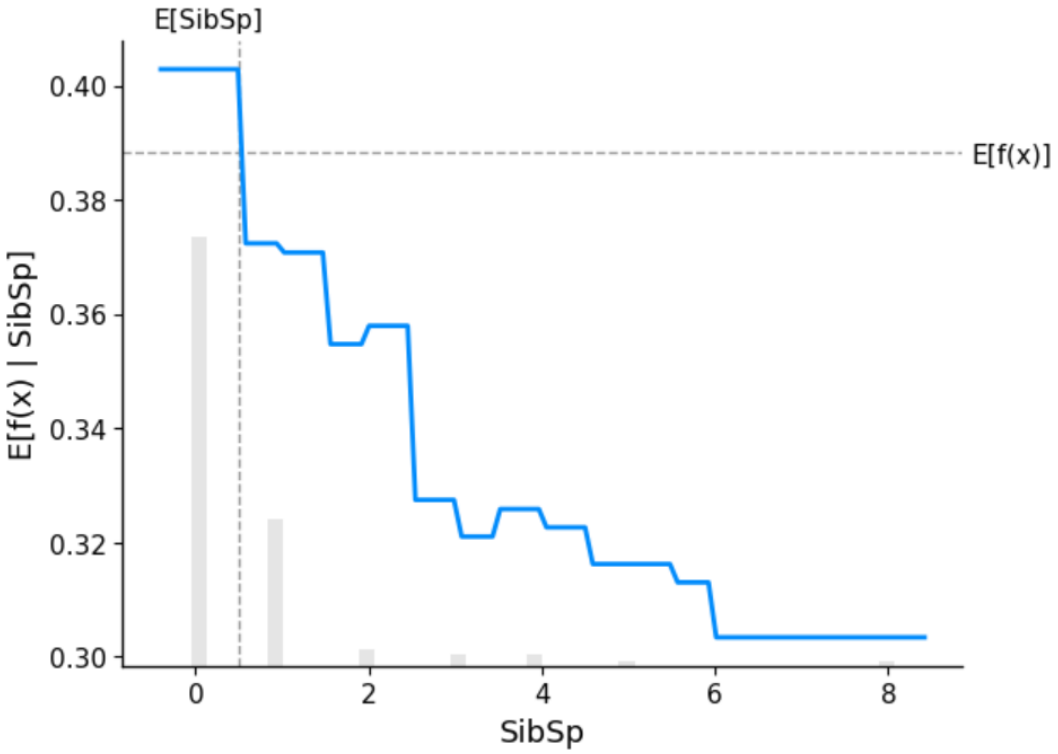
```
shap.partial_dependence_plot("SibSp", lm.predict, X_train, ice=False,
                             model_expected_value=True, feature_expected_value=True,)
```



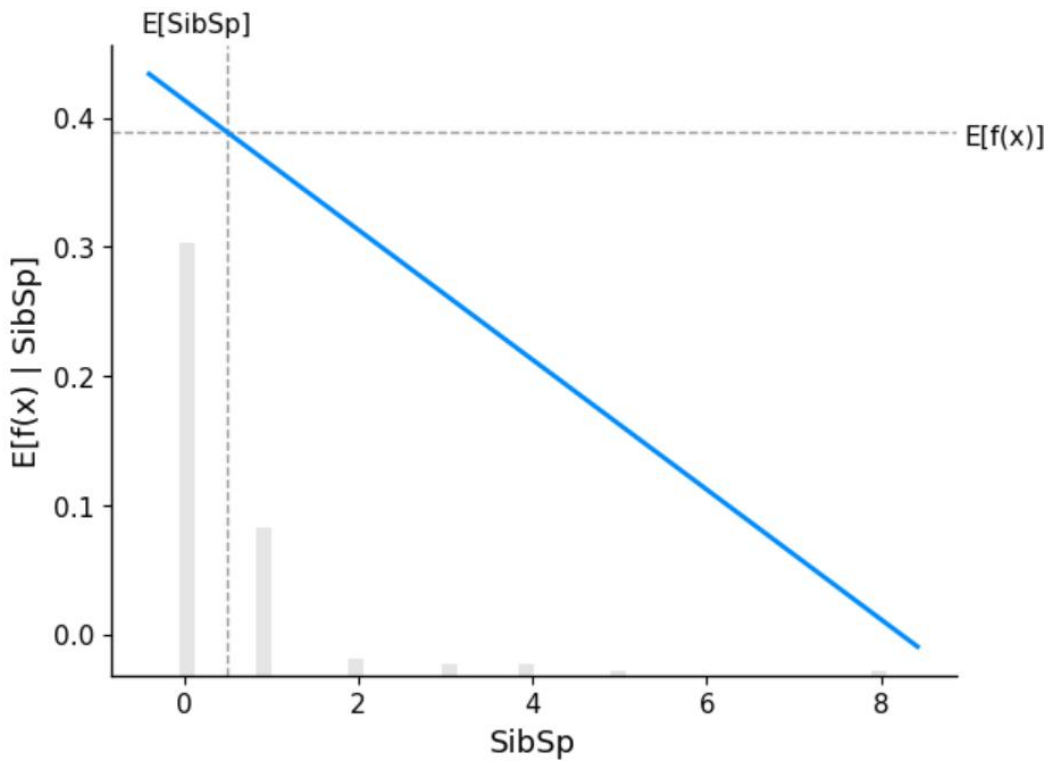
# SHAP Analysis – Feature Wise



**SibSp** – Random Forest vs. Linear Regression




Random Forest

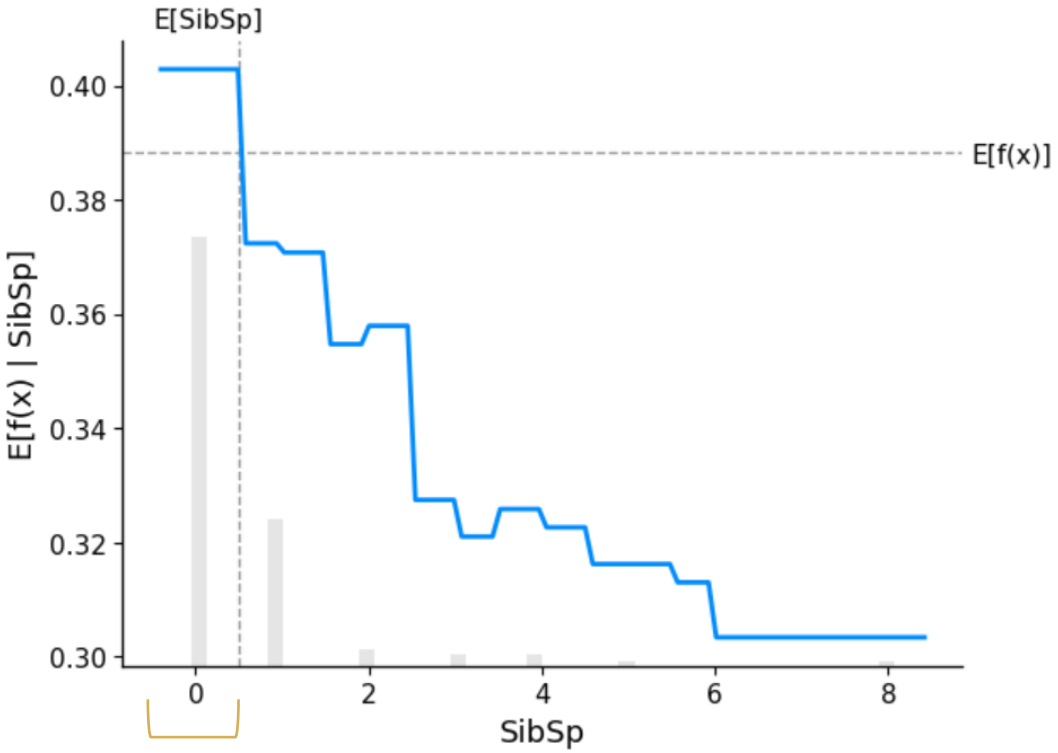



Linear Regression

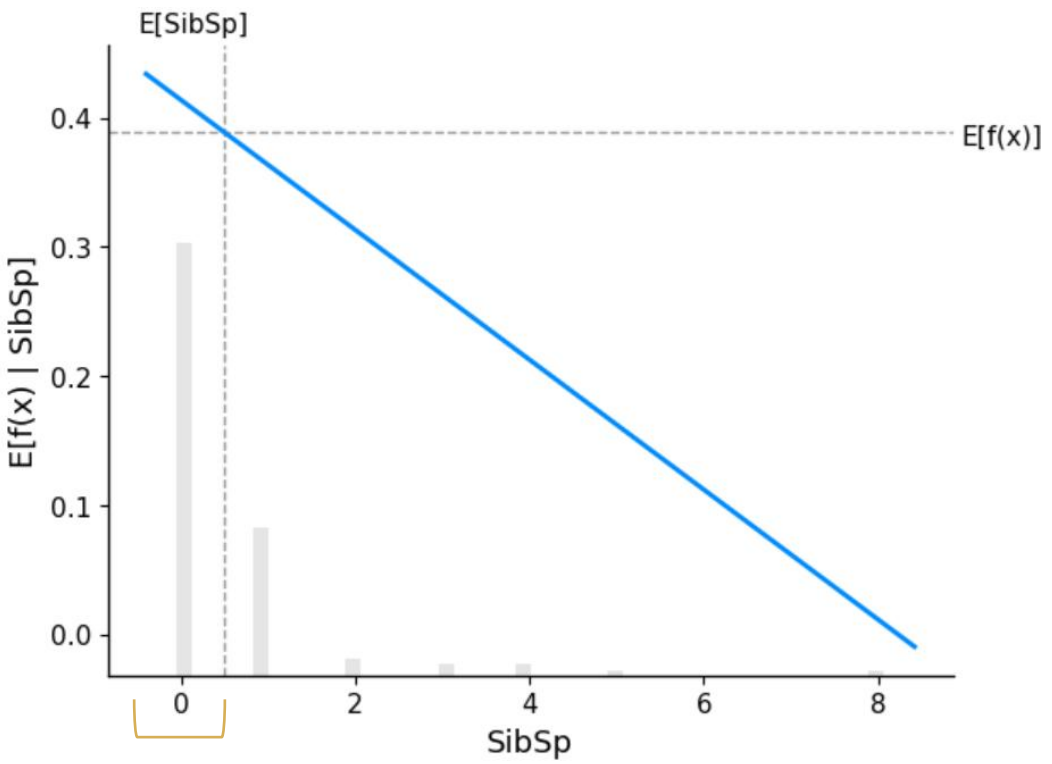
# SHAP Analysis – Feature Wise



**SibSp** – Random Forest vs. Linear Regression




Random Forest

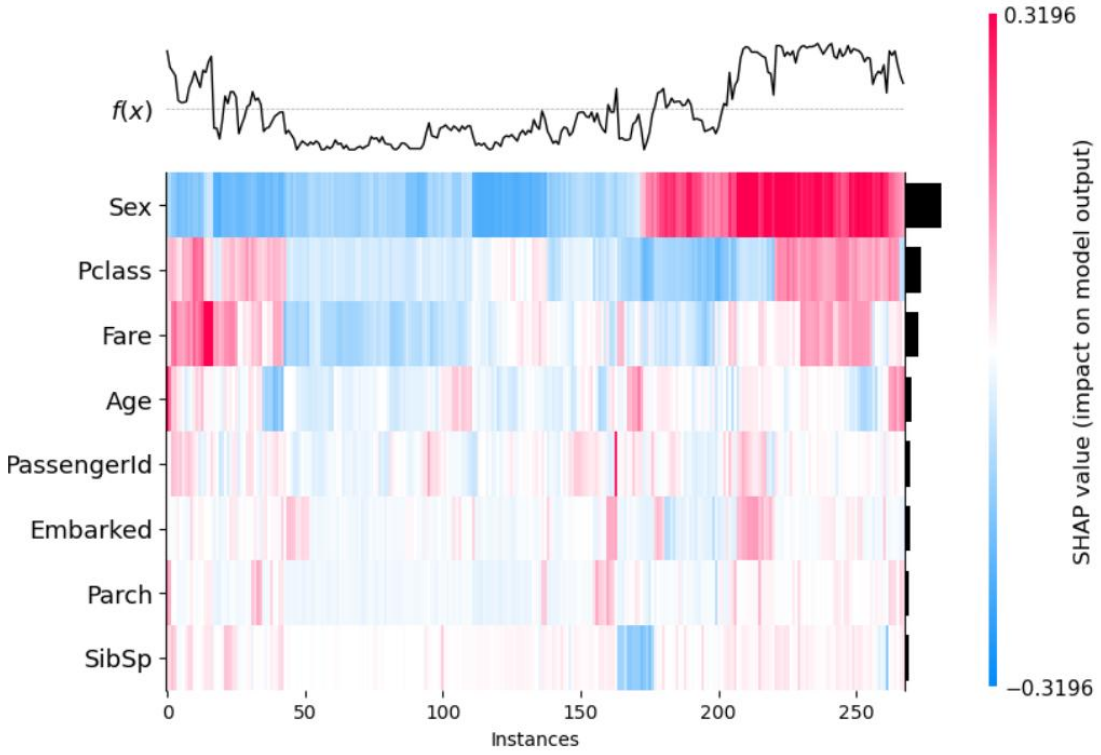



Linear Regression

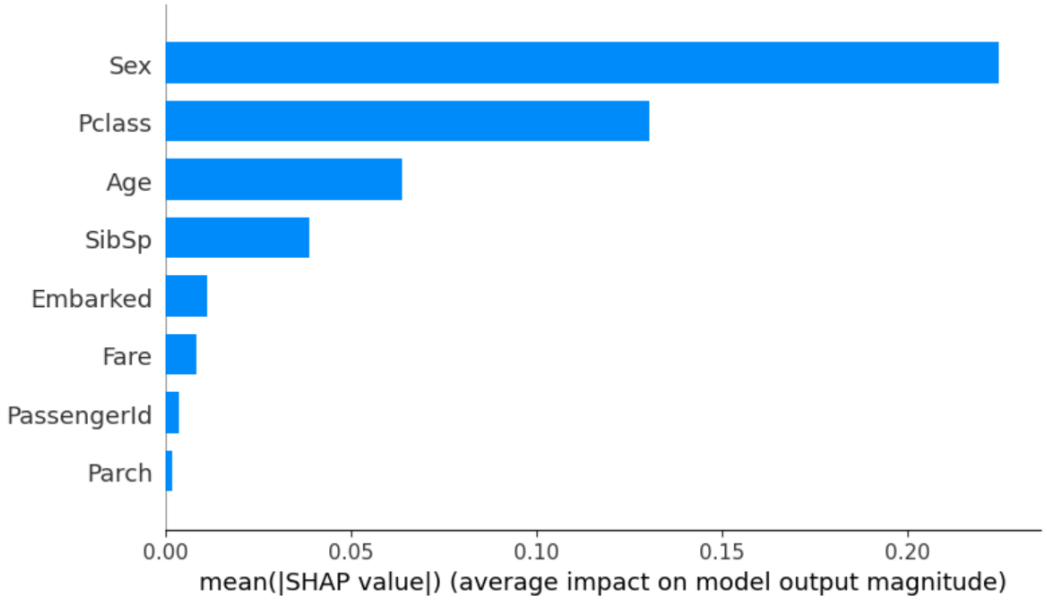
# SHAP Analysis – Random Forest vs. Linear Regression



Which features are more relevant to the model?




Random Forest

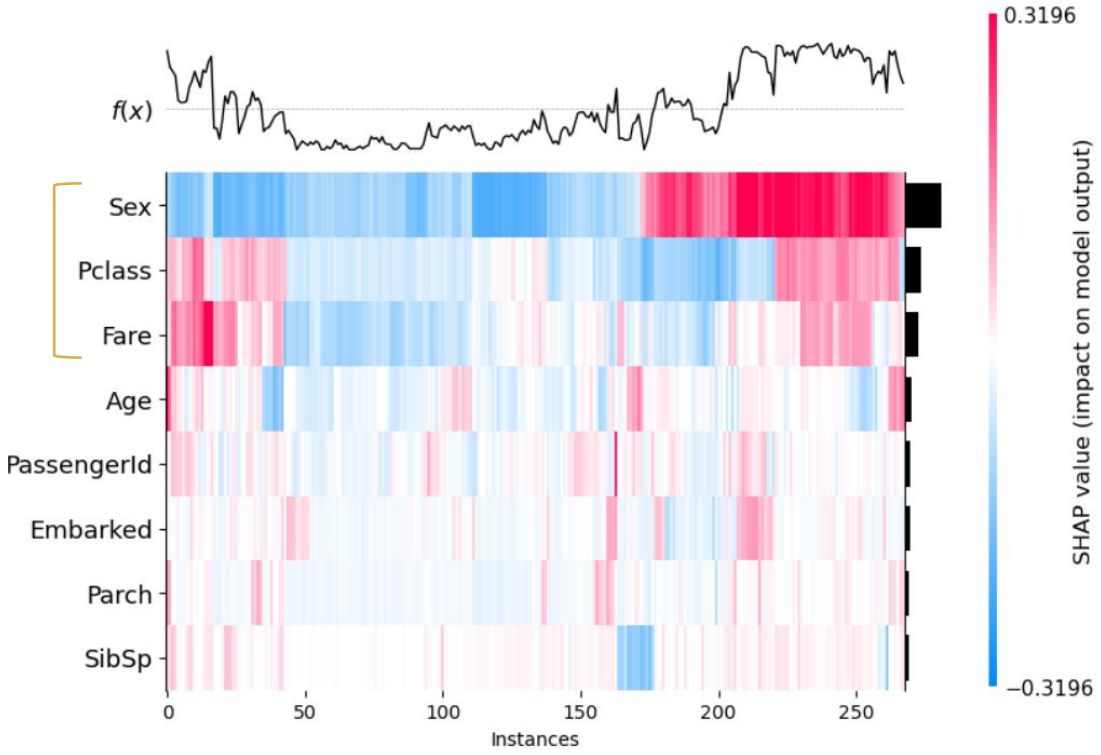



Linear Regression

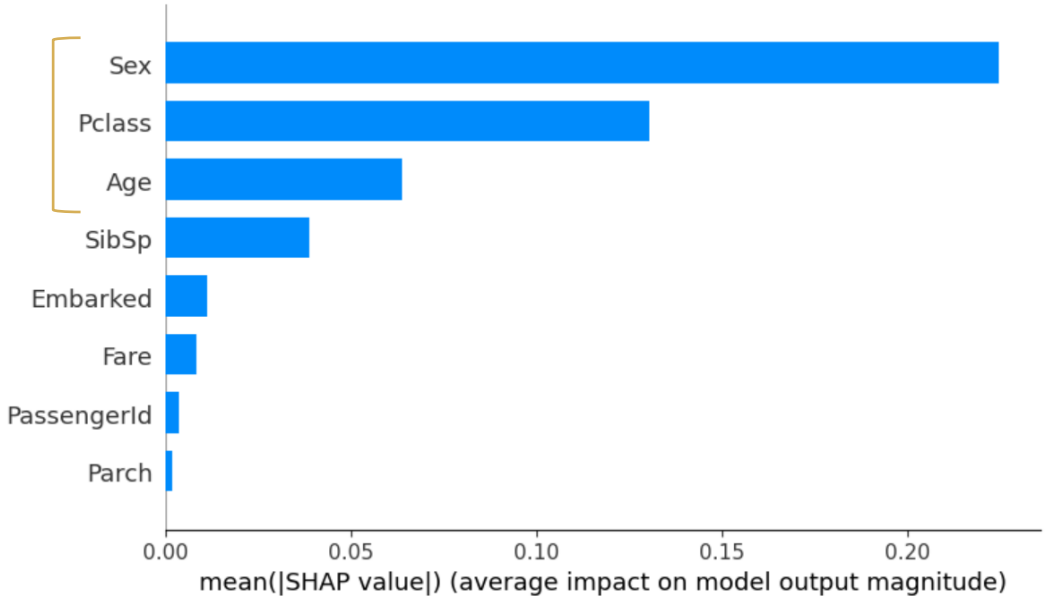
# SHAP Analysis – Random Forest vs. Linear Regression



Which features are more relevant to the model?




Random Forest




Linear Regression

# Hands On